Split Complex Lorentzian Dynamics

Part XXI: The Quaternionic Double-Cover Identification, Completed and Certified

The interrupted Task-21 reconstruction, Task-22 completion and full-project certification, and the return to local decomposition in Task XXIII

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Abstract

Part XXI originally documented an interrupted Task 21 run. The interruption was late and sharply localized: thirteen Task-21 Lean modules comprising 2655 lines had been written; twelve contained no `sorry`; all twenty-four unresolved placeholders, on twenty-two source lines, were confined to `NormalizedCentralCarrier.lean`. Before the timeout, the source had already packaged the intrinsic core as unit quaternions, constructed an explicit equivalence with $SU(2)$, identified the visible group with $SO(3)$, compared the intrinsic projection pointwise with quaternionic conjugation, recovered the kernel $\{\pm 1\}$, reformulated the global no-go as the nonexistence of a continuous section, identified the continuous centre with $\mathbb{C}^\times$, and written the full carrier as a central product.

Task 22 was deliberately restricted to completion and certification. It resumed at the exact timeout boundary, replaced the twenty-four placeholders by proofs, added only private helper lemmas needed for those proofs, built the three downstream modules without modifying them, created the missing Task-21 endpoint and a Task-22 completion marker, ran the focused Task-21 build, ran the complete project build, scanned the entire Task-21 directory for prohibited proof escapes, and audited the axioms of the principal endpoints. No intended Task-21 theorem was weakened, corrected, removed, or strengthened. The twelve previously source-closed Task-21 modules are byte-for-byte identical to the interrupted archive; only `NormalizedCentralCarrier.lean` changed, and the two endpoint files are new.

The certification is complete. The pre-timeout dependency chain builds in 8187 Lake jobs, the completed Task-21 endpoint builds in 8192 jobs, and the whole project builds in 8264 jobs under Lean 4.28.0 and Mathlib revision `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. The Task-21 directory contains zero `sorry`, `admit`, custom axioms, `implemented_by`, `native_decide`, `bv_decide`, `unsafe`, or `partial`. Seventy-eight distinct declarations are covered by `#print axioms`; every one reports exactly `[propext, Classical.choice, Quot.sound]`, with `sorryAx = 0`. The full build emits 114 style/linter warnings, all from pre-existing code; none comes from the files written by Task 22.

The formally certified group-level picture is therefore substantially stronger than the interrupted-paper status suggested. The project now contains abstract multiplicative-group equivalences

$$\mathcal{L} \simeq \mathbb{H}_1 \simeq SU(2), \qquad G_{\mathrm{vis}} \simeq SO(3),$$

together with a certified comparison of the intrinsic projection with quaternionic rotation and kernel $\{\pm 1\}$. It also contains

$$\mathcal{C}^\times \simeq \mathbb{C}^\times, \qquad \mathcal{L}_Z \simeq (\mathbb{C}^\times \times \mathbb{H}_1)/\Delta\mathbb{Z}_2,$$

the polar split $\mathbb{C}^\times \simeq \mathbb{R}_{>0} \times U(1)$, the normalized quotient $(U(1) \times \mathbb{H}_1)/\Delta\mathbb{Z}_2$, an abstract group equivalence of that normalized carrier with $U(2)$, and hence an abstract group equivalence

$$\mathcal{L}_Z \simeq \mathbb{R}_{>0} \times U(2).$$

Established mathematics identifies $\mathbb{H}_1 \cong SU(2) \cong Spin(3)$ and $(Spin(3) \times U(1))/\mathbb{Z}_2 \cong Spin^c(3) \cong U(2)$ [4, 7, 6, 10, 14, 12, 11]. The project has nevertheless not formalized named $Spin(3)$ or $Spin^c(3)$ library models, and only the core quaternion equivalence currently carries topology. No principal frame bundle or bundle lift has been reconstructed, so a spin structure itself remains beyond the present result.

Task 23 should therefore resume the project's decomposition/reconstruction method rather than extend the comparison catalogue. Its narrow target is the already certified double-cover map itself: decompose it into local sections/trivializations over a controlled open cover, derive the transition factors on overlaps, compare their $\{\pm 1\}$ data with the previously reconstructed bridge and integer-label structure, and test whether the total core carrier can be reconstructed from those local pieces. This is the smallest new step that moves from structure groups and a covering homomorphism toward bundle-level geometry without assuming a spin structure in advance.

Keywords: split-complex Lorentz structure; unit quaternions; $SU(2)$; $SO(3)$; $Spin(3)$; quaternionic double cover; central product; $\mathbb{C}^\times$; $U(2)$; $Spin^c(3)$ comparison; local sections; transition functions; formal verification; Lean 4; Conservation of Difficulty.

Contents

1 One paper for two runs: reconstruction and certification

Part XX contracted the broad translation problem into a finite list of group-theoretic identification tests [13, 1]. Task 21 attacked those tests and reached most of them before its execution budget expired. The first edition of Part XXI therefore had to distinguish source-level mathematical declarations from verified project endpoints. That distinction was essential at the time, because the archive contained no final `Task21.lean`, no Task-21 audit, no focused build report, no whole-project build report, and no axiom inventory [2].

Task 22 was designed specifically to remove that uncertainty and nothing else. It did not open a successor mathematics problem. Its acceptance condition was binary: either the interrupted Task-21 mathematics could be completed and made to survive a full project build, or Task 21 would remain incomplete. The completion archive now supplies the missing certification layer [3].

Methodological principle 1 (Merged-run reporting rule)**.** Task 21 and Task 22 are two execution phases of one mathematical episode. Task 21 supplied the substantial reconstruction; Task 22 supplied the missing proofs, endpoint packaging, and certification. They are therefore reported together in Part XXI rather than as two parallel papers about the same object.

2 Exact two-run provenance

The interrupted archive and the completion archive play different roles.

Item	Interrupted Task 21	Task 22 completion
Task-21 modules	13 modules, 2655 lines	15 files in final Task-21 chain, 3396 lines including endpoint files
Open proof holes	24 `sorry` in one module	0
Final Task21 endpoint	absent	present, 220 lines
Completion marker	absent	`Task22.lean`, 42 lines
Focused build	absent	PASS, 8192 jobs
Whole-project build	absent	PASS, 8264 jobs
Axiom audit	absent	78 distinct declarations audited
Task-21 audit	absent	present after green builds
Mathematical scope	identification reconstruction	completion/certification only

The completion audit records Lean 4.28.0 and Mathlib revision

`8f9d9cff6bd728b17a24e163c9402775d9e6a365`

with Lean commit

`7e01a1bf5c70fc6167d49c345d3bf80596e9a79b`

The completion archive supplied to this manuscript has SHA-256

`4ea45904b73e0fa8a0d52e7ff5d8e14a609ddae10c9f83699cc2be003860dac3`.

The audit also compares Git blob hashes between the interrupted and final source trees. Every inherited Task-21 file except `NormalizedCentralCarrier.lean` is byte-for-byte unchanged. This fact is unusually important: the successful certification does not come from rewriting the difficult source after the timeout. It comes from completing exactly the one file that had contained the placeholders.

3 The exact timeout boundary, now closed

The final Task-21 import chain is

$$\begin{aligned}&\texttt{SafeBase} \to \texttt{GroupPackaging} \to \texttt{CoreQuaternionEquiv} \to \texttt{SU2Model}\\ &\to \texttt{VisibleRotationCandidate} \to \texttt{CoreProjection} \to \texttt{SectionObstruction}\\ &\to \texttt{CentralComplex} \to \texttt{CentralProduct} \to \boxed{\texttt{NormalizedCentralCarrier}}\\ &\to \texttt{SignStructures} \to \texttt{TorsorPlacement} \to \texttt{NegativeControls} \to \texttt{Task21} \to \texttt{Task22}.\end{aligned} \tag{1}$$

In the interrupted archive the boxed module contained all twenty-four proof holes. Task 22 first rebuilt the nine modules ending at `CentralProduct`; they passed without any repair in 8187 jobs. This independently established that the source before the timeout boundary already typechecked in the inherited environment.

Task 22 then completed the fourteen originally unresolved constructions:

`Declaration`	Intended obligation
`polarEquiv`	$\mathbb{C}^\times \simeq \mathbb{R}_{>0} \times U(1)$
`polarEquiv_fst`, `polarEquiv_snd`	exact modulus and phase projections
`CUnit1_subset_CUnit`	normalized centre lies in the invertible centre
`CUnit1G`	group structure of the unit-modulus centre
`LiftZ1G`	group structure of the normalized full carrier
`posCentral`	positive real central homomorphism
`liftZSplit`	split $\mathcal{L}_Z \simeq \mathbb{R}_{>0} \times \mathcal{L}_{Z,1}$
`muNorm`	normalized product map from $U(1) \times \mathbb{H}_1$
`muNorm_surjective`	surjectivity onto $\mathcal{L}_{Z,1}$
`ker_muNorm`	diagonal sign kernel
`nuU2`	standard matrix homomorphism to $U(2)$
`nuU2_surjective`	surjectivity onto $U(2)$
`ker_nuU2`	equality of the normalized and matrix kernels

Two composite declarations, `normalizedQuotientEquiv` and `normalizedU2Equiv`, had contained no local placeholder in the interrupted archive but could not become valid endpoints until these dependencies were closed. They now build.

Status 1 (Timeout boundary resolved)**.** The timeout did not occur while recognizing the quaternion core, constructing $SU(2)$ or $SO(3)$, comparing the projection, computing the core kernel, identifying the complex centre, or deriving the full central product. It occurred only in the subsequent modulus/phase normalization and $U(2)$ comparison. Task 22 closes exactly this boundary.

4 Certified core: unit quaternions and $SU(2)$

Task 20 had already discovered the explicit quaternion map. Task 21 packages it as a multiplicative equivalence

$$\texttt{liftQuatEquiv} : \qquad \mathbb{H}_1 \simeq_{\mathrm{Grp}} \mathcal{L}, \tag{2}$$

and also as a homeomorphism of the underlying carriers

$$\texttt{liftQuatHomeo} : \qquad \mathbb{H}_1 \simeq_{\mathrm{Top}} \mathcal{L}. \tag{3}$$

The group equivalence is then composed with an explicit 2×2 complex matrix model of the unit quaternions. The resulting declarations `quatSUEquiv` and `liftSUEquiv` certify

$$\mathbb{H}_1 \simeq_{\mathrm{Grp}} SU(2), \qquad \mathcal{L} \simeq_{\mathrm{Grp}} SU(2). \tag{4}$$

This is no longer a comparison inferred from familiar formulas. It is a built project theorem. Classical mathematics then supplies the conventional name

$$\mathbb{H}_1 \cong SU(2) \cong Spin(3) \tag{5}$$

for the same compact group [4, 7, 14]. The project does not yet formalize the final equality with a named Mathlib three-dimensional `spinGroup`; that remains a comparison statement, not a gap in the certified $SU(2)$ identification.

5 Certified visible group and quaternionic projection

The visible transformations act on the reconstructed three-dimensional spatial carrier. Task 21 proves that the corresponding matrices preserve the Euclidean form, have determinant $+1$, and exhaust the standard three-dimensional special orthogonal group. The endpoint `visibleRotationEquiv` therefore certifies

$$G_{\text{vis}} \simeq_{\text{Grp}} SO(3). \tag{6}$$

The core projection is then compared pointwise with quaternionic conjugation on the imaginary quaternion space. The endpoint `task21_projection_diagram` certifies commutativity of the reconstructed projection with the standard quaternionic rotation action, while `task21_quat_rotation_kernel` and `task21_core_kernel` certify

$$\ker(\text{proj}\,|_{\mathcal{L}}) = \{+1, -1\}. \tag{7}$$

Surjectivity is separately exposed by `task21_projection_surjective`.

At the level of abstract groups and the explicitly compared action, the project has therefore reconstructed the conventional quaternionic double-cover diagram. In established terminology the same classical map is

$$Spin(3) \cong SU(2) \longrightarrow SO(3), \qquad \ker = \mathbb{Z}_2, \tag{8}$$

where the projection is quaternionic conjugation [7, 5].

Proof boundary 1 (What is formal and what is conventional naming)**.** *The formal project theorem names the upper group as `Lift` and, via an explicit equivalence, as $SU(2)$; the lower group is explicitly equivalent to $SO(3)$; the projection is compared pointwise with quaternionic rotation. The additional sentence "this is the standard $Spin(3) \to SO(3)$ cover" uses the established classical identification $SU(2) \cong Spin(3)$. No separate named three-dimensional Mathlib `spinGroup` equivalence is claimed.*

6 The global obstruction is now a certified section obstruction

Earlier tasks discovered the global obstruction in several languages: globally exact families, primitive bridges, relation completeness, and continuous rate assignments. Task 20 reduced it to the impossibility of a continuous quotient-compatible representative choice. Task 21 packages that statement directly at the group projection.

The endpoint layer now certifies:

1. a set-theoretic section of the core projection exists;
2. quotient-compatible choices and sections are equivalent in the precise implemented sense;
3. no continuous global section exists.

Thus

$$\exists s : G_{\mathrm{vis}} \to \mathcal{L}, \quad \mathrm{proj} \circ s = \mathrm{id}, \qquad \nexists\, s \text{ continuous with the same property.} \tag{9}$$

The importance of (9) is conceptual. The global obstruction is not the nonexistence of internal representatives. It is the impossibility of choosing them continuously across the full visible group. Once (8) is recognized, this is exactly the kind of obstruction expected from a nontrivial double covering. The project reached this conclusion from its earlier bridge/rate/relation reconstruction rather than importing covering theory as the starting point.

7 The continuous centre and the full central product

The double-cover core is not the whole internal carrier. Task 21 separately identifies the exact continuous centre. The explicit coefficient map

$$a\,1 + bS \longleftrightarrow a + ib$$

is packaged as

$$\mathcal{C}^{\times} \simeq_{\mathrm{Grp}} \mathbb{C}^{\times}. \tag{10}$$

The project also proves intrinsically

$$\mathcal{C}^{\times} \cap \mathcal{L} = \{+1, -1\}. \tag{11}$$

Let μ denote multiplication of a central unit and a core element. Task 21 proves that μ is surjective onto $\mathcal{L}_Z$ and has precisely the diagonal sign kernel. Consequently

$$\mathcal{L}_Z \simeq_{\mathrm{Grp}} (\mathbb{C}^{\times} \times \mathbb{H}_1)/\Delta\mathbb{Z}_2. \tag{12}$$

This result explains why the project carried both a two-valued sign and a continuous central bridge. The sign is the intersection of the quaternionic core with a larger complex multiplicative centre; it is not the whole centre.

8 What Task XXII adds: normalization and $U(2)$

Task 22 completes the stage at which the interrupted run stopped. Its first new certified equivalence is the polar decomposition

$$\texttt{polarEquiv} : \qquad \mathbb{C}^{\times} \simeq_{\mathrm{Grp}} \mathbb{R}_{>0} \times U(1). \tag{13}$$

The proof is explicit: a nonzero complex number is sent to its positive modulus and normalized phase, and the inverse multiplies radius and phase.

The normalized central subgroup and normalized full carrier are then packaged as groups. A positive central homomorphism and an explicit splitting map produce

$$\texttt{liftZSplit} : \qquad \mathcal{L}_Z \simeq_{\mathrm{Grp}} \mathbb{R}_{>0} \times \mathcal{L}_{Z,1}. \tag{14}$$

On the normalized factor, multiplication gives a surjective homomorphism

$$\mu_1 : U(1) \times \mathbb{H}_1 \longrightarrow \mathcal{L}_{Z,1} \tag{15}$$

with diagonal kernel

$$\ker \mu_1 = \Delta\mathbb{Z}_2. \tag{16}$$

Hence

$$\mathcal{L}_{Z,1} \simeq_{\mathrm{Grp}} (U(1) \times \mathbb{H}_1)/\Delta\mathbb{Z}_2. \tag{17}$$

Finally, the completed matrix map `nuU2` is proved surjective with the same kernel. The quotient therefore yields

$$\texttt{normalizedU2Equiv}: \qquad \mathcal{L}_{Z,1} \simeq_{\mathrm{Grp}} U(2). \tag{18}$$

The final endpoint `liftZU2Split` composes (14) and (18):

$$\boxed{\mathcal{L}_Z \simeq_{\mathrm{Grp}} \mathbb{R}_{>0} \times U(2).} \tag{19}$$

Methodological caveat 1 (Abstract group equivalence only)**.** *The audit is explicit that `liftSUEquiv`, `visibleRotationEquiv`, `liftZSplit`, `normalizedU2Equiv`, and `liftZU2Split` are equivalences of abstract multiplicative groups. Except for `liftQuatHomeo`, Task 21/22 does not formalize these as homeomorphisms, Lie-group isomorphisms, or smooth equivalences.*

Classically,

$$\mathit{Spin}^c(3) = (\mathit{Spin}(3) \times U(1))/\mathbb{Z}_2 \cong U(2) \tag{20}$$

is standard [6, 10, 12, 11]. The formal project has proved the quotient shape and the $U(2)$ group equivalence appearing in (20), but the installed library has no directly used named $\mathit{Spin}^c(3)$ object. The audit therefore correctly records the name $\mathit{Spin}^c(3)$ as comparison-level mathematics rather than as a formal declaration.

9 The post-timeout modules are now certified

The interrupted Task-21 archive already contained three downstream files with no local proof holes, but they could not be certified because they imported the unfinished normalization module. Task 22 builds them without modification.

9.1 Two different two-valued structures

`SignStructures.lean` proves that the invisible core sign and direction reversal are not the same object. The core transformation

$$g \longmapsto -g$$

is invisible under the projection, while direction reversal

$$n \longmapsto -n$$

is coupled to parameter reversal. The reversal quotient has more than two elements. Thus the $\mathbb{Z}_2$ kernel of the double cover must not be conflated with the antipodal direction quotient.

9.2 Half-odd labels as central phase weights

`TorsorPlacement.lean` places the previous rate torsor inside the certified complex centre. In the implemented complex coordinate, the bridge associated with rate b and parameter θ satisfies

$$\mathrm{toComplex}(\mathrm{labelBridge}(b, \theta)) = \exp(ib\theta). \tag{21}$$

The source further proves that equality after a full turn is equivalent to an integer difference of weights, while the full-turn sign retains only parity-type information. Hence the earlier $\mathbb{Z} + \frac{1}{2}$ torsor is not accidental notation: it becomes a concrete central phase-weight structure, and its reduction to $\{\pm 1\}$ loses information.

9.3 Negative controls

`NegativeControls.lean` remains intact and now builds. It includes noninjectivity of the core projection, explicit noncommuting visible transformations, and a counterexample showing that a two-element kernel does not by itself determine a homomorphism. These controls are important because the successful standard identification must follow from the explicit action and maps, not merely from matching dimensions or kernel cardinality.

10 Task XXII formal certification

The Task-22 run was intentionally judged by engineering criteria as strict as the mathematical ones.

10.1 Focused and full builds

The audit records three decisive build stages:

$$\begin{aligned} \texttt{lake build RequestProject.NullSectorTask21.CentralProduct} &\quad\Rightarrow\quad \text{PASS (8187 jobs)},\\ \texttt{lake build RequestProject.NullSectorTask21.Task21} &\quad\Rightarrow\quad \text{PASS (8192 jobs)},\\ \texttt{lake build} &\quad\Rightarrow\quad \textbf{PASS} \text{ (8264 jobs)}. \end{aligned}$$

The final build reports zero errors. No source outside `RequestProject/NullSectorTask21/` was modified by Task 22.

10.2 Proof-integrity scan

The complete fifteen-file Task-21 directory was scanned. The final counts are

Token / escape	occurrences
`sorry`	0
`admit`	0
custom `axiom`	0
`implemented_by`	0
`native_decide`	0
`bv_decide`	0
`unsafe`	0
`partial`	0

Thus the exact placeholder transition is

$$24 \longrightarrow 0. \tag{22}$$

10.3 Axiom audit

`Task21.lean` executes 69 `#print axioms` commands and `Task22.lean` executes 16, covering 78 distinct declarations after overlaps are removed. Every audited declaration reports exactly

$$\texttt{[propext, Classical.choice, Quot.sound]}.$$

No additional axiom occurs and `sorryAx = 0`.

10.4 Warnings

The whole project build emits 114 warnings. They are style/linter/deprecation warnings from pre-existing source: 4 in Task 11, 68 in Task 14, and 42 in Task 21 files written before Task 22. No warning is emitted by `NormalizedCentralCarrier.lean`, `Task21.lean`, or `Task22.lean`. The pre-existing Task-21 warning sources were deliberately not cosmetically rewritten because the Task-22 scope froze already source-closed files.

10.5 Failbuild provenance

The completion audit preserves three explicitly enumerated failed builds of `NormalizedCentralCarrier.lean` before the successful one. They were all proof-script/engineering failures: a carrier/subtype mismatch and equality orientation issue, an ambiguous `star_smul` plus a guessed coercion lemma, followed by two arithmetic cleanup failures around `field_simp`. None changed a theorem statement. The final module build passed in 8188 jobs.

There is one minor documentation inconsistency in the audit: the introductory sentence says "all four failures", but the ledger explicitly enumerates Failure 1, Failure 2, and Failure 3 before the passing build. The paper follows the enumerated evidence and records this as a counting typo, not as an additional mathematical or build failure.

Status 2 (Certification verdict)**. TASK XXI COMPLETION: PASS. TASK XXI FOCUSED BUILD: PASS. WHOLE PROJECT BUILD: PASS.**

11 Task XXII repair ledger: what was actually completed

The completion was not a wholesale rewrite. The repair ledger makes that visible declaration by declaration. The classification *proof completion* below means that the statement was inherited unchanged from the interrupted archive and only its proof hole was filled.

Declaration	Completion method
`polarEquiv`	Explicit forward map $z \mapsto (\|z\|, z/\|z\|)$ and inverse $(r, c) \mapsto r\,c$; both inverse identities and multiplicativity proved.
`polarEquiv_fst`, `polarEquiv_snd`	Coordinate formulas reduce to the constructed equivalence and close by definitional equality.
`CUnit1_subset_CUnit`	Unit modulus excludes simultaneous vanishing of the two central coordinates.
`CUnit1G`	Identity, multiplication and inverse closure; multiplication uses the two-square identity and inversion uses $zc(a, -b)$.
`LiftZ1G`	Identity, multiplication and inverse closure from the inherited central-swap law, normalized-centre closure and core inversion.
`posCentral`	Positive-real central elements $zc(r, 0)$ and their inverses are packaged without changing the declared positive-real model.
`liftZSplit`	A new helper homomorphism `splitMap` is shown injective and surjective. Existence uses $r = \sqrt{a^2 + b^2}$; uniqueness uses the intrinsic coordinate norm `nsq`.
`muNorm`	The existing map $(c, q) \mapsto \mathrm{ofComplex}(c) \star \mathrm{fromQuat}(q)$ is proved multiplicative. The quaternion core is not replaced by $SU(2)$ in the intrinsic construction.
`muNorm_surjective`	Surjectivity follows from the defining shape of the normalized carrier and the inherited quaternion inverse identities.
`ker_muNorm`	Kernel derived from the already proved intrinsic intersection $\mathcal{C}^\times \cap \mathcal{L} = \{\pm 1\}$, giving exactly the diagonal sign pair.

Declaration	Completion method
`nuU2`	The existing single matrix model is reused; unitary membership and homomorphism laws are proved.
`nuU2_surjective`	For $A \in U(2)$ choose a unit-modulus square root of $\det A$, remove that phase to obtain an $SU(2)$ element, then use the inherited $SU(2)$ surjectivity. No dimension argument is used.
`ker_nuU2`	Determinant forces the phase square to one; injectivity of the quaternion matrix model finishes the kernel computation and identifies it with `muNorm.ker`.
`normalizedQuotientEquiv`	Statement and source left untouched; it becomes valid once its kernel and surjectivity dependencies are completed.
`normalizedU2Equiv`	Statement and source left untouched; it becomes valid once `nuU2` and its kernel/surjectivity are complete.

Task 22 added private helper lemmas for coordinate norms, central coefficients, normalized membership, packaging maps, and the matrix-kernel calculation. These helpers introduce no new object of study. The audit classifies the fourteen original holes as proof completion and the endpoint files as packaging/refactor only; there are no theorem weakenings and no theorem strengthenings.

Status 3 (Mathematical stability across the timeout)**.** No intended Task-21 theorem turned out to be false or overstrong. The transition from the interrupted archive to the certified archive changes proof status, not mathematical statement boundaries.

12 Certified result ledger

The merged Task-21/22 status is now:

Claim	Final status
$\mathcal{L} \simeq$ unit quaternions	CERTIFIED
Core maps also form a homeomorphism	CERTIFIED
$\mathcal{L} \simeq SU(2)$	CERTIFIED, abstract group
$G_{\mathrm{vis}} \simeq SO(3)$	CERTIFIED, abstract group
Intrinsic projection = quaternionic rotation action	CERTIFIED
Core kernel = $\{\pm 1\}$	CERTIFIED
Set-theoretic section exists	CERTIFIED
No continuous global section	CERTIFIED
$\mathcal{C}^\times \simeq \mathbb{C}^\times$	CERTIFIED
$\mathcal{C}^\times \cap \mathcal{L} = \{\pm 1\}$	CERTIFIED
$\mathcal{L}_Z \simeq (\mathbb{C}^\times \times \mathbb{H}_1)/\Delta\mathbb{Z}_2$	CERTIFIED, abstract group
$\mathbb{C}^\times \simeq \mathbb{R}_{>0} \times U(1)$	CERTIFIED, abstract group
$\mathcal{L}_Z \simeq \mathbb{R}_{>0} \times \mathcal{L}_{Z,1}$	CERTIFIED, abstract group
$\mathcal{L}_{Z,1} \simeq (U(1) \times \mathbb{H}_1)/\Delta\mathbb{Z}_2$	CERTIFIED, abstract group

Claim	Final status
$\mathcal{L}_{Z,1} \simeq U(2)$	CERTIFIED, abstract group
$\mathcal{L}_Z \simeq \mathbb{R}_{>0} \times U(2)$	CERTIFIED, abstract group
Core sign distinct from direction reversal	CERTIFIED
Half-odd labels become central phase weights	CERTIFIED
Full turn equality $\leftrightarrow$ integer weight difference	CERTIFIED
Full-turn sign loses integer-weight information	CERTIFIED
Named $Spin(3)$ model in Lean	NOT FORMALIZED
Named $Spin^c(3)$ model in Lean	NOT FORMALIZED
Topological/Lie equivalences beyond `liftQuatHomeo`	NOT FORMALIZED
Principal spin structure over a base	NOT CONSTRUCTED
w_2 obstruction	NOT CONSTRUCTED
Physical spin degree of freedom	NOT DERIVED

13 How close is the reconstruction to a spin structure now?

The completion makes the hierarchy much sharper than it was in Parts XIX and XX.

13.1 Level I: the spin-group carrier

At project level,

$$\mathcal{L} \simeq SU(2)$$

is now certified. At conventional mathematical level, $SU(2) \cong Spin(3)$. The reconstruction has therefore reached the correct group carrier for three-dimensional spin geometry.

13.2 Level II: the double-cover homomorphism

The lower group is certified as $SO(3)$ and the projection is compared pointwise with quaternionic rotation, with kernel $\{\pm 1\}$ and surjectivity. Thus the project has also reached the group-theoretic covering homomorphism underlying

$$Spin(3) \to SO(3).$$

This is substantially stronger than observing a $2\pi/4\pi$ sign behavior.

13.3 Level IIb: the normalized central extension

The larger carrier now yields

$$\mathcal{L}_{Z,1} \simeq U(2)$$

and intrinsically

$$\mathcal{L}_{Z,1} \simeq (U(1) \times \mathbb{H}_1)/\Delta\mathbb{Z}_2.$$

Classically this is the low-dimensional quotient used to define $Spin^c(3)$. The project is therefore also very close to the standard $Spin^c(3)$ group environment, although the conventional name is not itself a formal project object.

13.4 Level III: a spin structure

A spin structure is not merely the existence of $Spin(3)$ and a double-cover homomorphism. It requires a base space, an oriented orthonormal frame-type principal $SO(n)$ bundle over that base, and a compatible lift of that bundle through the spin covering map [8, 9]. None of those bundle-level objects has yet been reconstructed from the project data.

The correct conclusion is therefore:

> The group-theoretic substrate of three-dimensional spin geometry is now certified, and the normalized continuous extension matches the standard $U(2)$ group appearing in $Spin^c(3)$.
>
> A spin structure itself has not yet been reconstructed.

14 What the completion resolves from Parts XIX and XX

Part XIX found a cluster of apparently separate phenomena: local half-odd labels, integer differences, central bridges, a $2\pi/4\pi$ internal sign, an antipodal direction involution, relation completeness, and a global no-go. It was not yet possible to determine whether they belonged to one standard object or several coupled ones.

Task 20 supplied the first fixed anchor by identifying the core with unit quaternions. Task 21 then wrote the expected $SU(2)$, $SO(3)$, projection, centre, and central-product identifications. Task 22 certifies that these declarations actually coexist in one green formal development and completes the $U(2)$ normalization.

Several earlier near-identifications are therefore no longer merely thematic:

1. the quaternionic core and visible rotations are linked by an explicit certified projection;
2. the invisible sign is exactly the kernel of that projection;
3. the no-global-choice result is a certified no-continuous-section statement;
4. the continuous bridge sector is exactly a complex multiplicative centre;
5. the full internal carrier is a central product of that centre with the quaternionic core;
6. the half-odd/integer rate structure becomes a concrete phase-weight structure inside the centre;
7. normalizing the centre gives the standard $U(2)$ abstract group carrier.

At the same time, Task 21 preserves an important non-identification: the direction-reversal $\mathbb{Z}_2$ is not the core-kernel $\mathbb{Z}_2$. The project therefore does not collapse every two-valued structure into the same sign.

15 Conservation of Difficulty after certification

The broad explanatory detour of Parts XIX–XXII is now largely closed at group level. The project no longer needs to ask whether the accumulated algebra resembles spinorial structures. It has explicit certified groups, maps, kernels, quotients, and a section obstruction.

The remaining difficulty has moved one categorical level upward. A group covering is not yet a bundle lift over a physical or geometric base. The next question should therefore not be another broad search for names. It should ask whether the already reconstructed local exact representatives and overlap data are the local pieces of the certified covering object.

This is a particularly natural return to the original decomposition philosophy. Instead of starting with a textbook principal bundle and mapping the project into it, the next task can decompose the certified global projection into local data and ask whether the former `Dom(v)`, bridge, and label structures reappear as its transition system.

Methodological principle 2 (Return to reconstruction)**.** Task 23 should not assume a spin structure. It should reconstruct the local geometry of the certified double cover from the project's own domains and overlaps. Only after that local structure has been recovered should a later task ask whether an independent frame-type bundle over a base is present and can be lifted.

16 Outlook: Task XXIII as a narrow local decomposition/reconstruction task

Task 23 should begin from the green Task-21 endpoint produced by Task 22. It should introduce no new physical dynamics and no new standard target group. Its object of study is the already certified surjective projection

$$p : \mathcal{L} \longrightarrow G_{\mathrm{vis}}, \qquad \ker p = \{\pm 1\}. \tag{23}$$

The task should be divided into an intrinsic reconstruction phase and a late comparison phase.

16.1 Phase A: decompose the certified projection locally

The first objective is to construct a controlled family of open subsets $V_i \subseteq G_{\mathrm{vis}}$ and local maps

$$s_i : V_i \longrightarrow \mathcal{L}, \qquad p \circ s_i = \mathrm{id}_{V_i}, \tag{24}$$

using the project's existing parameter domains and exact local representatives wherever possible.

The domains should not be chosen merely because a standard covering theorem guarantees their existence. Task 23 should first test whether the earlier `Dom(v)` construction, pushed through the exact visible quotient, already supplies suitable local sections or can be refined minimally to do so.

Required questions are:

1. Which earlier direction/parameter domains descend to open subsets of G_{vis} on which a continuous local section exists?
2. Can a small explicit cover be constructed from those domains?
3. Are the local sections unique up to the certified kernel sign once a normalization is fixed?

16.2 Phase B: derive overlap transitions rather than assume them

On each nonempty overlap $V_i \cap V_j$, define the relative factor by

$$t_{ij}(x) = s_i(x)^{-1} s_j(x). \tag{25}$$

Task 23 should prove directly that

$$t_{ij}(x) \in \ker p = \{\pm 1\}. \tag{26}$$

It should then determine the strongest regularity statement available. Because the target is discrete, continuity should force local constancy on connected overlap components, but this must be proved from the actual implemented topology if topology is formalized sufficiently; otherwise the result must be stated at the strongest justified level.

On triple overlaps, derive the transition law

$$t_{ij} t_{jk} = t_{ik} \tag{27}$$

from the section definitions. Do not call the collection a cocycle before the exact coefficient group and overlap law have been formalized.

16.3 Phase C: compare with the earlier bridge and label data

Task 23 should then return to the pre-identification reconstruction and ask:

1. Is the sign transition t_{ij} exactly the reduction of the earlier integer overlap-rate difference to its full-turn sign?

2. Which information in the $\mathbb{Z}$ torsor is lost when one passes to $\{\pm 1\}$ transitions?

3. Does the earlier primitive bridge factor through the certified continuous centre before normalization and through the kernel sign after restriction to the core cover?

4. Is Task 19 relation completeness exactly the condition that these local core sections are compatible with every visible coincidence relation?

These are equivalence tests, not assumptions.

16.4 Phase D: reconstruct the total core from local pieces

If the local sections and transition laws are obtained, Task 23 should attempt one reconstruction theorem: glue the local products

$$V_i \times \{\pm 1\}$$

using the derived transition maps and construct an explicit map from the resulting quotient/glued carrier to `Lift`. Prove an equivalence at the strongest available level.

This is the key reconstruction test. It asks whether the global quaternionic core can be recovered from the local section data that arose earlier in the project, rather than merely recognizing the global core after the fact.

16.5 Late comparison only

Only after the intrinsic local reconstruction is closed may Task 23 identify the resulting object with the standard principal $\mathbb{Z}_2$ covering associated with $SU(2) \to SO(3)$. This late comparison may use established covering/principal-bundle terminology.

Task 23 must stop there.

Proof boundary 2 (Task-23 scope). *Task 23 must not construct a spacetime frame bundle, a principal $SO(1,3)$ bundle, a spin structure, a Spin^c structure over a manifold, a Stiefel–Whitney class, or a physical spin state. Its sole new mathematical object is the* local decomposition and reconstruction of the already certified core double cover*. That is narrow enough to preserve the project's anti-target-bias method and strong enough to prepare the next genuine bundle-level question.*

16.6 Task-23 acceptance boundary

A focused Task 23 should be considered successful if it resolves the following finite chain, without broadening beyond it:

1. construct at least one explicit open cover of the visible group on which continuous local sections of p are proved to exist;

2. derive the overlap transition maps and prove that their values lie in the certified kernel $\{\pm 1\}$;

3. prove the triple-overlap composition law and determine the strongest justified local-constancy statement;

4. compare the sign transitions with the earlier integer overlap differences and state exactly what information the sign reduction forgets;

5. either reconstruct `Lift` from the local products and transitions, or isolate the first missing hypothesis that prevents the reconstruction.

A negative result at any one of these points is acceptable if it supplies an explicit counterexample or a precise missing hypothesis. The task should not compensate for such a failure by opening a wider bundle or cohomology search.

The intended dependency DAG is therefore

$$\begin{aligned} \text{certified } p : \mathcal{L} \to G_{\mathrm{vis}} &\longrightarrow \text{ local sections} \\ &\longrightarrow \{\pm 1\} \text{ overlap transitions} \\ &\longrightarrow \text{ bridge/label comparison} \\ &\longrightarrow \text{ local-to-global reconstruction of } \mathcal{L}. \end{aligned} \tag{28}$$

Only after the final arrow is closed should standard principal-cover terminology be admitted in a comparison-only layer.

17 Conclusion

The first edition of Part XXI had to stop at a verification boundary. The mathematical source was already substantial, but an interrupted Lean run is not a certified theorem layer. Task 22 was therefore deliberately uncreative: finish exactly the missing file, preserve the existing source boundary, build everything, expose the endpoints, audit the axioms, record the failed proof scripts, and stop.

That strategy succeeded. No source-closed Task-21 theorem had to be changed. The placeholder count fell from twenty-four to zero. The focused Task-21 build passed in 8192 jobs, the entire project passed in 8264 jobs, and all seventy-eight audited declarations have exactly the established project axiom baseline. The formerly tentative Task-21 identifications are now project results.

At group level the reconstruction has reached a remarkably specific classical structure:

$$\mathcal{L} \simeq SU(2), \qquad G_{\mathrm{vis}} \simeq SO(3), \qquad \ker p = \{\pm 1\},$$

with the intrinsic action identified with quaternionic rotation, together with the larger central extension

$$\mathcal{L}_Z \simeq (\mathbb{C}^\times \times \mathbb{H}_1)/\Delta\mathbb{Z}_2 \simeq \mathbb{R}_{>0} \times U(2)$$

as abstract multiplicative groups. The half-odd labels are simultaneously located as central phase weights, while the project proves that direction reversal is a different two-valued structure from the kernel sign.

This is now much closer to the mathematical infrastructure underlying spin geometry than the early $2\pi/4\pi$ heuristic ever justified. But the remaining gap is precise: no base-space frame bundle and no lift of such a bundle have been reconstructed. The project has the relevant structure groups and covering homomorphism, not yet a spin structure.

That precision makes a normal reconstruction task possible again. Task 23 should decompose the certified double cover into local sections and transition factors, compare those factors with the project's earlier domains, bridges, and integer labels, and then attempt to reconstruct the global core from the local pieces. It is a small categorical move upward, but it returns to the central methodological question of the project: not merely whether a familiar object can be recognized at the end, but whether its dependency structure can be recovered from the primitive constraints that produced it.

A Final Task-21/22 source inventory

Module	Final lines	Role
`SafeBase`	112	reconstruction coordinates and spatial lemmas
`GroupPackaging`	321	packaged core, centre, full carrier, projections
`CoreQuaternionEquiv`	218	unit-quaternion equivalence and homeomorphism
`SU2Model`	162	explicit 2×2 complex model and $SU(2)$ equivalence
`VisibleRotationCandidate`	549	$SO(3)$ action, orientation, surjectivity
`CoreProjection`	130	quaternion-conjugation comparison, kernel
`SectionObstruction`	138	section existence and continuous-section no-go
`CentralComplex`	189	$\mathcal{C}^\times \simeq \mathbb{C}^\times$
`CentralProduct`	255	intersection, product kernel, full quotient
`NormalizedCentralCarrier`	614	polar split, normalized quotient, $U(2)$ equivalence; completed by Task 22
`SignStructures`	147	kernel sign versus direction reversal
`TorsorPlacement`	154	phase weights and integer/full-turn structure
`NegativeControls`	145	anti-overclaim controls
`Task21`	220	endpoint packaging and axiom audit
`Task22`	42	completion marker and unresolved-declaration axiom checks
Total	**3396**	

B Task-22 build and integrity ledger

Check	Result
Pre-timeout dependency build through `CentralProduct`	PASS, 8187 jobs
Completed `NormalizedCentralCarrier` build	PASS, 8188 jobs after recorded proof-script repairs
Focused `Task21` build	PASS, 8192 jobs
Whole-project `lake build`	PASS, 8264 jobs
Whole-project errors	0
Warnings	114, pre-existing; 0 from Task-22-written files
Task-21 `sorry` count	0
Task-21 prohibited proof escapes	0
Distinct declarations in axiom audit	78
Axioms for every audited declaration	`propext`, `Classical.choice`, `Quot.sound`
`sorryAx`	0

Check	Result
Theorem weakenings/corrections/removals	0
Inherited source-closed files modified	0 of 12

C Formal claim boundary after Task XXII

The following remain intentionally outside the certified Task-21/22 theorem layer:

1. a named formal equivalence with Mathlib's general Clifford-algebra `spinGroup`;
2. a named formal $Spin^c(3)$ object;
3. topological or smooth group equivalences for $SU(2)$, $SO(3)$, $U(2)$ and the central-product decompositions beyond the specific `liftQuatHomeo`;
4. any principal frame bundle over a base space;
5. a spin or $Spin^c$ structure on such a bundle;
6. a w_2 obstruction or classification;
7. any physical spin observable, state, statistic, dynamics, or interaction.

These exclusions are not defects in the present theorem. They mark the next categorical layers that have not yet been reconstructed.

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